

Laboratory practice N°1. Impact test.

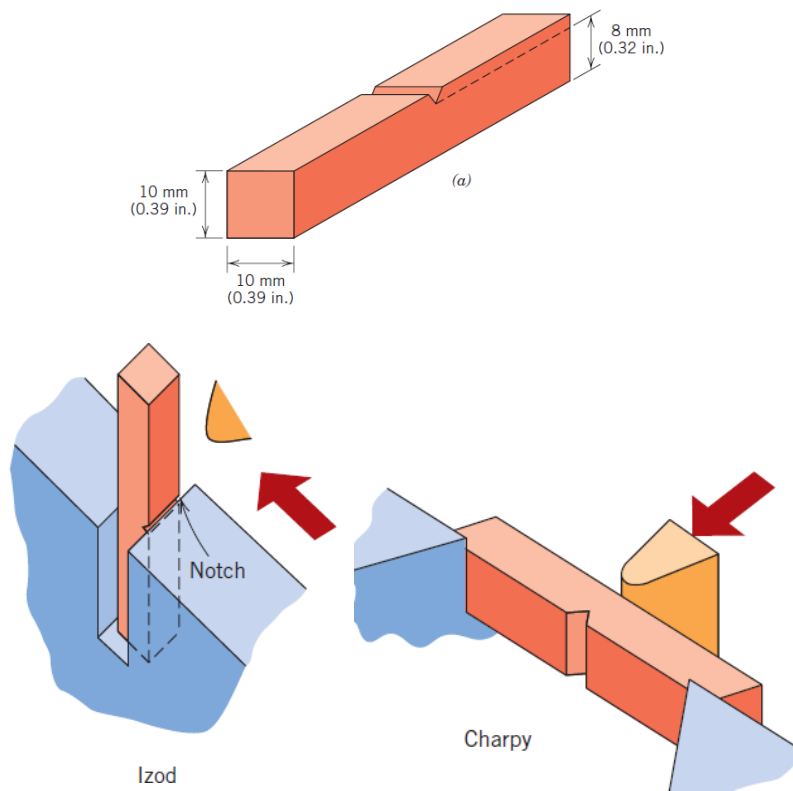
Objectives:

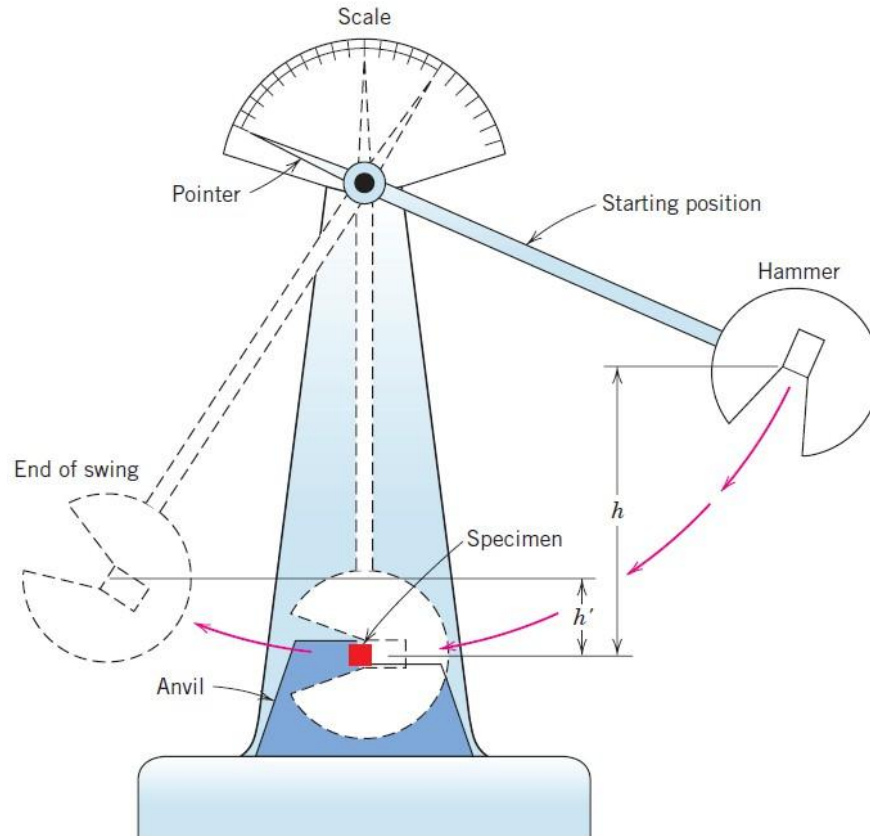
- Perform an Charpy impact test on sample of carbon steel and aluminum.
- Obtain the toughness for each material above.
- Compare energy absorption of the different materials above.

Part I. Measure of the toughness

Toughness is a measure of the ability of a material to absorb energy up to fracture. Specimen geometry as well as the manner of load application are important in toughness determination. For dynamic (high strain rate) loading conditions and when a notch (or point of stress concentration) is presented, notch toughness is assessed by using an impact test.

Two standardized tests, the Charpy (for metals) and Izod (for polymers), were designed to measure the impact energy. For both Charpy and Izod, the specimen is in the shape of a bar of square cross section, into which a V-notch could be machined (see figure below), in which case is called notch toughness. The machine for testing the samples is shown in the figure. The load is applied as an impact from a weighted pendulum hammer that is released from a certain position at a fixed height h . The specimen is positioned at the base as shown. Depending on the type of test: Charpy or Izod, the positioning of the sample differs. Upon release, a knife edge mounted on the pendulum strikes and fractures the specimen at the notch, which acts as a point of stress concentration for this high velocity impact blow. The pendulum continues its swing, rising to a maximum height h' , which is lower than h . The energy absorption, computed from the difference between h and h' is a measure of the impact energy.





Take a sample of each of the following materials:

- Carbon steel, 2Ps
- Aluminum, 2Ps

1) Measure the dimensions of each sample. Write the data taken in here.

Samples	dimensions
Carbon Steel	
Aluminum	

2) Mark the middle point on each sample.

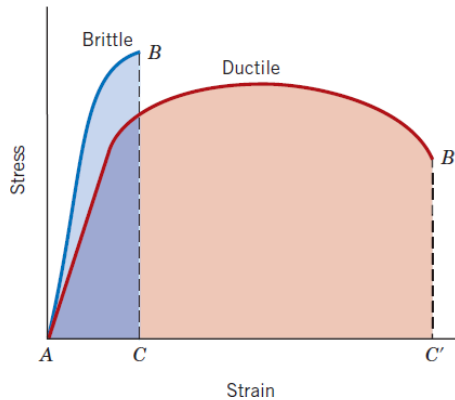
- 3) Place each of the samples in the machine and perform the Charpy impact test of each material. Do not forget to note the energy measured of each.
- 4) Place this sample in the machine and do the Charpy test. Do not forget to note the energy measured.
- 5) Give the following data for each tested sample:
 - a. Cross-sectional area
 - b. Energy measured
 - c. Resilience (J/cm^2)

Data of each sample	Value	
	Carbon Steel	Aluminum
Cross-sectional area		
Energy measured		
Resilience (J/cm^2)		

- 6) Classify the materials by increasing ductility.
- 7) Explain the difference between carbon steel, and aluminum sample.
- 8) Do you get the same resilience for PE with or without an initial notch? Explain.

Part II. Theoretical questions.

- 1) Looking at the following data of stress vs strain for brittle and ductile materials, which one has higher toughness? The one giving curve ABC or AB'C'?



- 2) Is ductility related to the toughness of the material?
- 3) From the previous graph, which material is the stiffest? Why?
- 4) Which is the strongest? Why?
- 5) Which will experience the greatest percent reduction in area? Why?